The Supply Side of HANK

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- Not empirically supported:
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- We look at the implications for labor supply, marginal costs, inflation dynamics, and monetary policy tradeoffs.

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  becoming & remaining HtM
- Policy implications: no divine coincidence and strict inflation targeting is sub-optimal.

literature

HANK Model

McKay, Nakamura, and Steinsson (2016) / one asset version of Kaplan, Moll, and Violante (2018)

$$V_t(b_{h,t}, z_{h,t}) = \max_{c_{h,t},\, b_{h,t+1},\, n_{h,t}} \left\{ \frac{c_{h,t}^{1-\sigma}}{1-\sigma} - \frac{n_{h,t}^{1+\varphi}}{1+\varphi} + \beta \sum_{z_{h,t+1}} \Pr(z_{h,t+1} \mid z_{h,t})\, V_{t+1}(b_{h,t+1}, z_{h,t+1}) \right\}$$

subject to

$$c_{h,t} + b_{h,t+1} = (1+r_t) b_{h,t-1} + W_t z_{h,t}\, n_{h,t} - \tau_t \bar{\tau}(z_{h,t}) + d_t \bar{d}(z_{h,t})$$

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- taxes and dividends proportional to $z_{h,t}$.

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- Substituting (1) into (2) yields:

$$N_t = W_t^{1/\varphi} \underbrace{\int z_{h,t}^{(1+\varphi)/\varphi} c_{h,t}^{-\sigma/\varphi} d\mathcal{F}_t(b,z)}_{\mathcal{K}_t\text{: cross-sectional moment}=C_t^{-\sigma/\varphi}\text{ in RANK}} . \tag{3}$$

The $\mathcal{Q}$ Wedge

$$N_t = W_t^{1/\varphi} C_t^{-\sigma/\varphi} \int z_{h,t}^{(1+\varphi)/\varphi} \left(\frac{c_{h,t}}{C_t} \right)^{-\sigma/\varphi} d\mathcal{F}_t(b,z) = N_t^R \mathcal{Q}_t^{-1/\varphi}, \tag{4}$$

$$N_t^R = W_t^{1/\varphi} C_t^{-\sigma/\varphi} \tag{5}$$

$$\mathcal{Q}_t = \left[\int z_{h,t}^{(1+\varphi)/\varphi} \left(\frac{c_{h,t}}{C_t} \right)^{-\sigma/\varphi} d\mathcal{F}_t(b,z) \right]^{-\varphi} = \mathcal{K}_t^{-\varphi} C_t^{-\sigma}. \tag{6}$$

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- $\mathcal{Q}_t$ is the ratio between the marginal utility of aggregate consumption and the average, productivity–adjusted marginal utility of individual consumption.
- $\mathcal{Q}_t = 1$ in RANK and "Canonical" HANK (Auclert, Rognlie, and Straub, 2025).

$\mathcal{Q}_t$ Decomposition

two sources of heterogeneity

- The composite wedge can be split into a productivity term, a consumption-inequality term, and a covariance term:

$$\mathcal{Q}_t^{-1/\varphi} = \underbrace{\mathcal{F}^z}_{E_h[z_{h,t}^{(1+\varphi)/\varphi}]} \cdot \underbrace{Q_{c,t}}_{E_{h,t}\left[\left(c_{h,t}/C_t\right)^{-\sigma/\varphi}\right]} + \underbrace{\mathcal{C}_t}_{\mathrm{Cov}_{h,t}\left(z_{h,t}^{(1+\varphi)/\varphi}, \left(c_{h,t}/C_t\right)^{-\sigma/\varphi}\right)} \tag{7}$$

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- Both $\mathcal{F}^z$ and $Q_{c,t}$ increase with cross-sectional dispersion in productivity and consumption, respectively.

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$$\mathcal{Q}_t = \exp\left(-\frac{1+\varphi}{2\varphi}\sigma_z^2 - \frac{\sigma(\sigma+\varphi)}{2\varphi}\sigma_{c,t}^2 + \frac{(1+\varphi)\sigma}{\varphi}\rho_{zc,t}\sigma_z\sigma_{c,t}\right), \tag{8}$$

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- For empirically relevant calibrations (Heathcote, Storesletten, and Violante (2014)), and always when $\sigma = \varphi = 1$, we get $\mathcal{Q}_t \leq 1$.

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- More on HANK model

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- A rise in inequality (lower $\hat{q}_t$) suppresses inflationary pressure at a given output gap: the outward shift of the AS curve reduces inflationary pressure.

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  - Limited risk sharing: within but not across types.

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  - Two household types (islands): savers (S) and hand-to-mouth households (H).
  - Homogeneous productivities.
  - Zero liquidity.
  - Limited risk sharing: within but not across types.
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Details

A Tractable HANK

Bilbiie (2025)

- $\mathcal{Q}_t^{-1/\varphi} = \mathcal{F}^z Q_{c,t} + \mathcal{C}_t$
  - $\frac{c_h}{C}$: per capita variation in consumption inequality (intensive margin)
  - $\mathcal{F}_t(b, z)$: cross sectional distribution of agents (extensive margin)

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Details

Idiosyncratic state & endogenous HtM share

- 2 states Markov chain with stochastic and cyclical transition probabilities.

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\begin{aligned}
p_{S,t} &= p_S \exp(\eta_S \hat{y}_t + e_{S,t}), \\
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Aggregate Labor Supply Wedge

- Q_t is the ratio between the *marginal utility of the average consumption* and the *average of the marginal utilities of consumption*.

$$Q_t \equiv \left[\left(\frac{C_{S,t}}{C_t} \right)^{1-\lambda_t} \left(\frac{C_{H,t}}{C_t} \right)^{\lambda_t} \right]^{\sigma} = \left(\frac{\Gamma_t^{\,1-\lambda_t}}{\lambda_t + (1-\lambda_t)\Gamma_t} \right)^{\sigma} \tag{15}$$

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- Intensive margin: changes in inequality $\Gamma_t \equiv \frac{C_{S,t}}{C_{H,t}}$
- Extensive margin: changes in agent shares λ_t
- Q_t measures the deviation from the frictionless marginal rate of substitution.

Aggregate Labor Supply Wedge

$$
\begin{aligned}
(1-\tau^w)W_t &= \overbrace{\exp(-\xi_t)\left(N_{S,t}^{\varphi}C_{S,t}^{\sigma}\right)^{1-\lambda_t}\left(N_{H,t}^{\varphi}C_{H,t}^{\sigma}\right)^{\lambda_t}}^{\text{Average } MRS_t} && (16)\\
&= \underbrace{\exp(-\xi_t)N_t^{\varphi}C_t^{\sigma}}_{\text{Frictionless } MRS_t^*} \cdot \underbrace{\left[\left(\frac{C_{S,t}}{C_t}\right)^{1-\lambda_t}\left(\frac{C_{H,t}}{C_t}\right)^{\lambda_t}\right]^{\sigma}}_{Q_t} && (17)
\end{aligned}
$$

First-Order Approximation (FOA): Supply Side

Steady state | Full model

no profits redistribution $\tau^D = 0$

$$\hat{c}_{S,t} = E_t\hat{c}_{S,t+1} - (1-p)E_t\hat{\gamma}_{t+1} - \frac{1}{\sigma}\left(\hat{\imath}_t - E_t\hat{\pi}_{t+1} + E_t\Delta\hat{\xi}_{t+1}\right) + p\frac{\Gamma^\sigma - 1}{\sigma}E_t\hat{p}_{S,t+1} \tag{18}$$

$$\hat{c}_{H,t} = \hat{w}_t + \hat{n}_t \tag{19}$$

$$\hat{q}_t = -\sigma(\lambda - \psi)\hat{\gamma}_t - \sigma(\lambda\gamma - \psi g)\hat{\lambda}_t \tag{20}$$

$$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} - \eta\hat{y}_t + p_H e_{H,t} - \frac{1-\lambda}{\lambda}p_S e_{S,t} \tag{21}$$

$$\hat{\gamma}_t = \hat{c}_{S,t} - \hat{c}_{H,t} \tag{22}$$

$$\widehat{mc}_t = \hat{w}_t - \hat{a}_t = -\hat{\xi}_t + \varphi\hat{n}_t + \sigma\hat{c}_t + \hat{q}_t - \hat{a}_t \tag{23}$$

$$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + \frac{\kappa}{\sigma + \varphi}\widehat{mc}_t. \tag{24}$$

- $\psi = \lambda\frac{C_H}{Y}$, $\gamma = \log\Gamma$, $g \equiv \Gamma - 1 > \gamma$, $p \equiv \frac{p_S}{p_S + (1-p_S)\Gamma^\sigma}$, $\eta \equiv \frac{1-\lambda}{\lambda}p_S\eta_S + p_H\eta_H$.

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- From (20) if $\Gamma = 1 \Rightarrow \gamma = 0$, $g = 0$, $\lambda = \psi$, then $\hat{q}_t = 0$ (Bilbiie (2025)).

Implications

- Shocks that worsen transition probabilities (lower p_S, higher p_H or both) have independent recessionary demand-side, deflationary supply-side, and worsening distributional effects.

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- Optimal monetary policy should overlean against these shocks.
- How much? There is a trade-off.
- Strict inflation targeting is sub-optimal (unlike Bilbiie (2025)).

Partial Equilibrium - Labor Market

- Partial profits redistribution case: $\tau^D = \psi = \lambda \frac{C_H}{Y} < \lambda$.

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- Lowercase letters without hats denote log levels.
- Preference shock $\hat{\xi}_t$ that triggers an increase in idiosyncratic risk $\hat{\lambda}_t$.

Partial Equilibrium - Labor Market

Employment

- The equilibrium is:

$$y_t = y_t^* = \frac{1+\varphi}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi} \hat{\xi}_t$$

details

Partial Equilibrium - Labor Market

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- Using the production function we get the demand determined employment:

$$n_t = y_t^* - a_t = \frac{1-\sigma}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi} \hat{\xi}_t$$

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$$n_t = y_t^* - a = \underbrace{\left(\frac{1+\varphi}{\sigma+\varphi} a - a\right)}_{y^* - a} + \frac{1}{\sigma+\varphi} \hat{\xi}_t.$$

details

Partial Equilibrium - Labor Market

Labor Supply

- Using the expression for wages we get the aggregate labor supply:

$$w_t = -\hat{\xi}_t + \varphi n_t + \sigma y_t + \hat{q}_t$$

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details

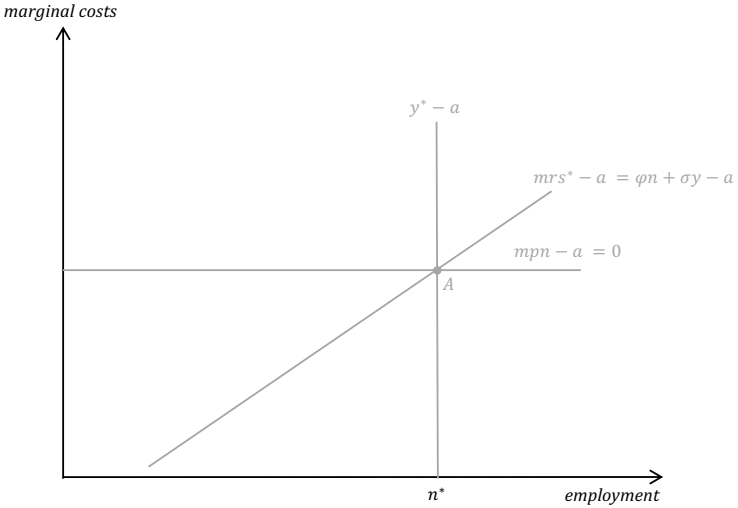
Partial Equilibrium - Labor Market

Equilibrium Markup

- Substituting the equilibrium values of n_t and y_t in the wage equation:

$$
\begin{aligned}
w_t &= -\hat{\xi}_t + \varphi\left(\frac{1-\sigma}{\sigma+\varphi}a + \frac{1}{\sigma+\varphi}\hat{\xi}_t\right) + \sigma\left(\frac{1+\varphi}{\sigma+\varphi}a + \frac{1}{\sigma+\varphi}\hat{\xi}_t\right) + \hat{q}_t \\
\Rightarrow \mu_{P,t} &= a - w_t = -\hat{q}_t
\end{aligned}
$$

Partial Equilibrium - Labor Market



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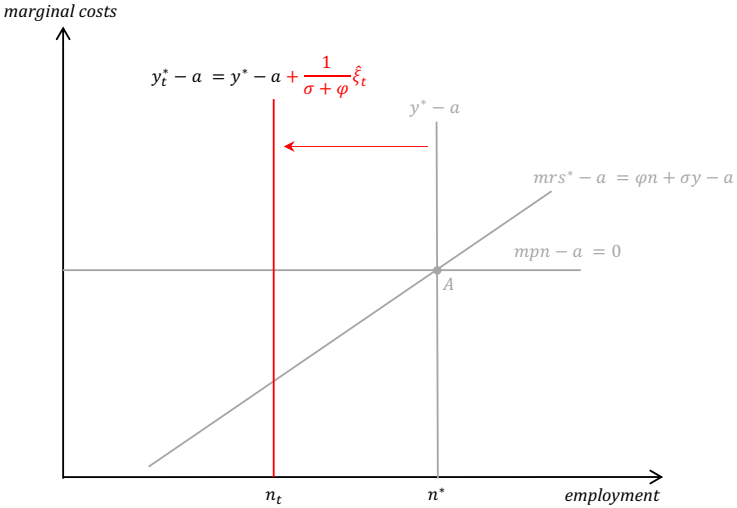

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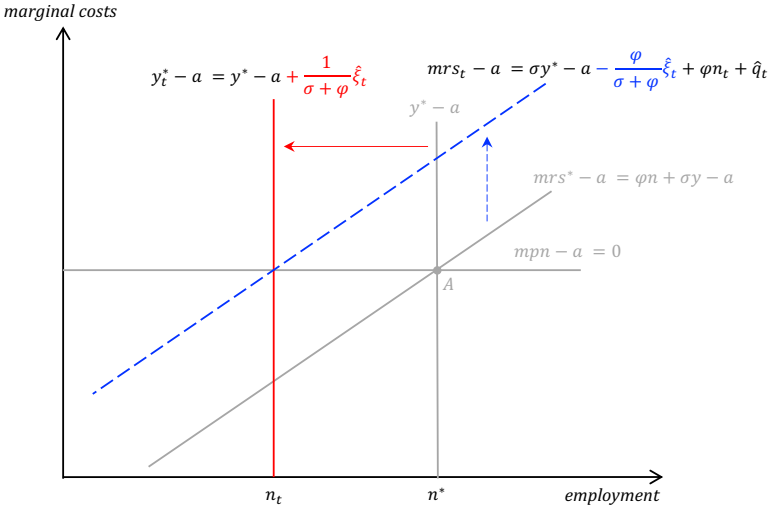

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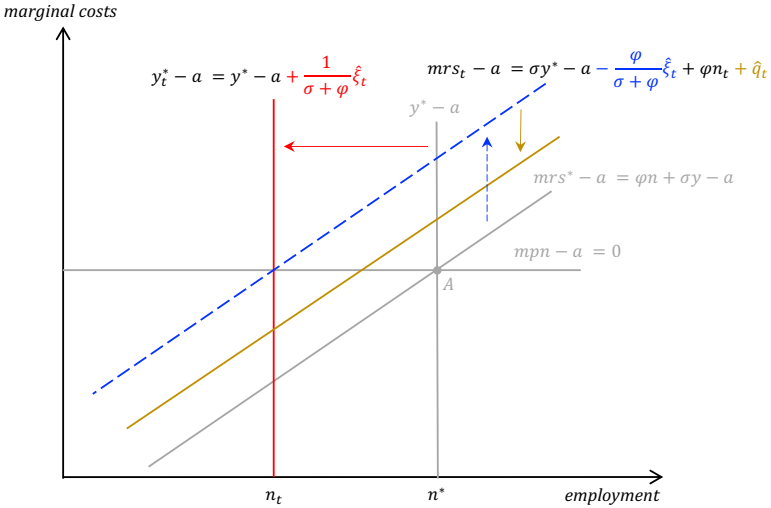

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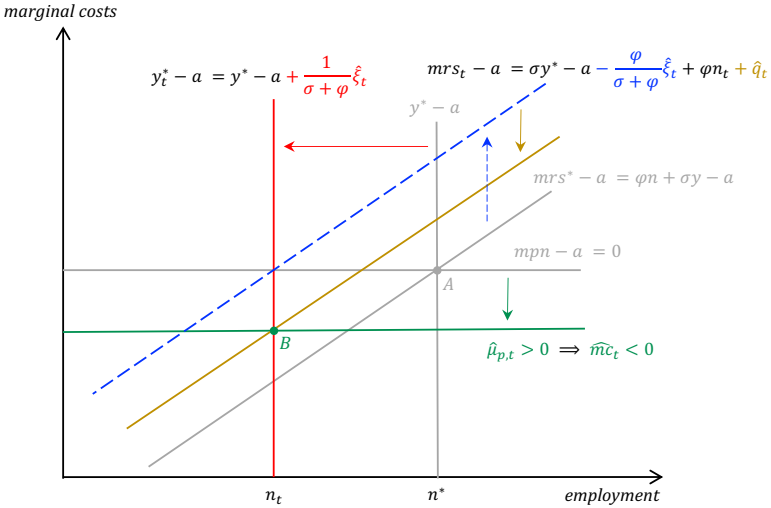

Takeaways

Partial Equilibrium - Labor Market

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- Idiosyncratic risk generates a trade-off for monetary policy via its transmission through the extensive margin of $\hat{q}_t$.
- The trade-off is between $\downarrow$ output gap and $\downarrow$ inflation.
- This happens outside of the ZLB.
- Preliminary results with heterogeneity in productivities in the T-HANK model suggest these shocks might also generate stagflation.

Next Steps

- THANK with heterogeneous productivity.

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- Q_t: sufficient-statistic cross-sectional wedge, which feeds into labor supply, marginal costs, and the Phillips curve.
- Decompose Q_t into two relevant margins: intensive and extensive margins of heterogeneity.
- Idiosyncratic risk shocks become recessionary, deflationary, and distributionally adverse.
- Policy should lean harder against these shocks, and strict inflation targeting is not optimal.

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Related Literature

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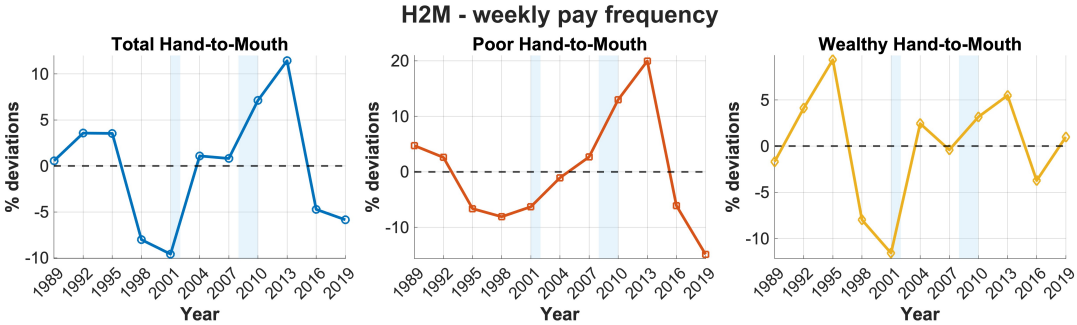
- Heterogeneity in consumption baskets & NKPC: Olivi, Sterk, and Xhani (2024), Neyer and Stempel (2025).
- Optimal policy and the divine coincidence in HANK: McKay and Wolf (2025), Dávila and Schaab (2023), Acharya, Challe, and Dogra (2023), Del Negro et al. (2025).

Evolution of HtM

Back



Fluctuations around the linear trend in the proportion of Hand-to-Mouth households, identified as in Kaplan and Violante (2014), from the Survey of Consumer Finances (SCF). A linear time trend has been removed to isolate cyclical variation. The blue shaded areas indicate NBER recessions.

- **Sizable** fluctuations around the trend
- **Countercyclical** H2M share dynamics

Empirical evaluation of $\mathcal{Q}_t$ Back

$$\mathcal{Q}_t = \exp\left(-\frac{1+\varphi}{2\varphi}\sigma_z^2 + \frac{(1+\varphi)\sigma}{\varphi}\rho_{zc,t}\sigma_z\sigma_c - \frac{\sigma(\sigma+\varphi)}{2\varphi}\sigma_c^2\right). \tag{25}$$

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- $\mathcal{Q}_t \leq 1$ if and only if the exponent is non-positive.

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- $\mathcal{Q}_t \leq 1$ if and only if the exponent is non-positive.
- Dividing through by $\sigma_z\sigma_c \cdot \sigma(1+\varphi)/\varphi > 0$ and defining $r_t \equiv \sigma_c/\sigma_z$, this reduces to:

$$\rho_{zc,t} \leq \frac{(1+\varphi) + \sigma(\sigma+\varphi)\, r_t^2}{2\sigma(1+\varphi)\, r_t}. \tag{26}$$

Empirical evaluation of $\mathcal{Q}_t$ Back

- Evaluate the implied values for $\mathcal{Q}$ for different calibration of σ and φ, calibrated at $\sigma_z^2 = 0.315$, $\sigma_c^2 = 0.118$, $\rho_{zc} = 0.52$ (Heathcote, Storesletten, and Violante, 2014).

Table: Implied values of $\mathcal{Q}_t$ from equation (25) for alternative calibrations of σ and φ.

$\sigma \setminus \varphi$	0.5	1	2	3	5	10
0.5	0.683	0.772	0.820	0.837	0.851	0.861
1	0.706	0.793	0.840	0.856	0.870	0.880
2	0.631	0.765	0.842	0.870	0.893	0.910
3	0.445	0.656	0.796	0.850	0.895	0.930
5	0.109	0.339	0.596	0.720	0.837	0.938
10	≈0	0.008	0.103	0.239	0.470	0.778

$1 - \mathcal{Q}_t$

Back

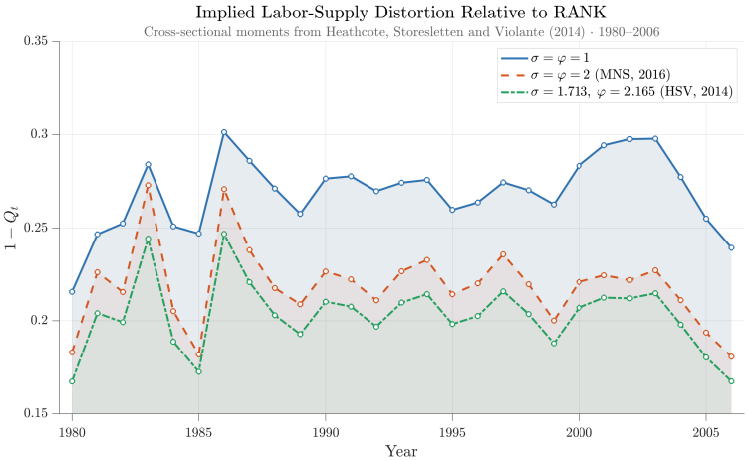

Figure: Fluctuations in the labor-supply distortion relative to RANK, for different calibrations.

HANK Model: Firms

Back

- The final-good firm aggregates differentiated intermediates with CES technology:

$$Y_t = \left(\int_0^1 y_{jt}^{1/\mu} \, dj \right)^{\mu}.$$

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- Price adjustment is costly à la Rotemberg:

$$\psi_t(p_{jt}, p_{jt-1}) = \frac{\mu}{\mu - 1} \frac{1}{2\kappa} \left[\log \left(\frac{p_{jt}}{p_{jt-1}} \right) \right]^2 Y_t.$$

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- In symmetric equilibrium, inflation satisfies

$$\log(1 + \pi_t) = \kappa \left(\frac{W_t}{A_t} - \frac{1}{\mu} \right) + \frac{1}{1 + r_{t+1}} \frac{Y_{t+1}}{Y_t} \log(1 + \pi_{t+1}),$$

while dividends are

$$d_t = Y_t - W_t N_t - \psi_t.$$

HANK Model: Policy

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- So monetary policy governs intertemporal prices, while fiscal policy passively balances the budget given the interest bill on government liquidity.

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- Goods market clearing allocates output between consumption and nominal rigidities:

$$Y_t = \underbrace{\sum_{z \in \mathcal{E}} \int c_t(z, b) \mathcal{F}_t(z, b)\, db}_{\text{Aggregate consumption}} + \psi_t.$$

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- These conditions close the model: savings equal public liquidity, labor clears in efficiency units, and output is split between consumption and price-adjustment costs.

HHs problem Back

- Utilitarian family head maximising the average lifetime utility across all agents

HHs problem

Back

- Utilitarian family head maximising the average lifetime utility across all agents
- Limited risk-sharing: *within* but not *across* types

$$\mathcal{U}_t = \int_0^1 u_{j,t} + \beta E_t \mathcal{U}_{t+1} = (1-\lambda_t) u_{S,t} + \lambda_t u_{H,t} + \beta E_t \mathcal{U}_{t+1} \tag{27}$$

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- Hand-to-mouth households receive a transfer that redistributes among them a share τ^D of aggregate profits D, financed through a lump-sum tax on savers.

HHs problem

Back

$$C_{S,t}^{-\sigma} = \beta(1+i_t)E_t\left\{\frac{\exp(\Delta\xi_{t+1})}{1+\pi_{t+1}}\left[p_{S,t+1}C_{S,t+1}^{-\sigma} + (1-p_{S,t+1})C_{H,t+1}^{-\sigma}\right]\right\} \tag{28}$$

$$C_{H,t} = (1-\tau^W)W_{H,t}N_{H,t} - T_{H,t} = (1-\delta\tau)W_tN_t + \frac{\tau^D}{\lambda_t}D_t \tag{29}$$

- $T_{H,t} = (\delta\tau - \tau^W)W_{H,t}N_{H,t} - \frac{\tau^D}{\lambda_t}D_t$ denotes the type-specific, per-capita lump-sum taxes net of transfers.

$$N_{j,t}^{\varphi}C_{j,t}^{\sigma}\exp(-\xi_t) = (1-\tau^W)W_{j,t}. \tag{30}$$

Summary of the Model

Back

Condition	Equation
1. Euler equation Savers	$C_{S,t}^{-\sigma} = \beta(1+i_t)E_t\left\{\frac{\exp(\Delta\xi_{t+1})}{1+\pi_{t+1}}\left[p_{S,t+1}C_{S,t+1}^{-\sigma} + (1-p_{S,t+1})C_{H,t+1}^{-\sigma}\right]\right\}$
2. Budget constraint H	$C_{H,t} = W_t N_t + \frac{\tau^D}{\lambda_t} D_t - \delta\tau W_t N_t$
3. Labor supply S	$N_{S,t}^{\varphi} C_{S,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{S,t}$
4. Labor supply H	$N_{H,t}^{\varphi} C_{H,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{H,t}$
5. Marginal cost	$MC_t = (1-\tau)\frac{W_t}{A_t}$
6. Aggregate labor	$N_t = N_{S,t}^{1-\lambda_t} N_{H,t}^{\lambda_t}$
7. Aggregate Wage	$W_t = W_{S,t}^{1-\lambda_t} W_{H,t}^{\lambda_t}$
8. Production function	$Y_t = A_t N_t$
9. Aggregate cons.	$Y_t = (1-\lambda_t)C_{S,t} + \lambda_t C_{H,t}$
10. Profits	$D_t = Y_t(1-MC_t) - (1-\delta)\tau W_t N_t = Y_t - (1-\delta\tau)W_t N_t$
11. Share dynamic	$\lambda_t = \lambda_{t-1} p_{H,t} + (1-\lambda_{t-1})(1-p_{S,t}) = (p_{S,t} + p_{H,t} - 1)\lambda_{t-1} + 1 - p_{S,t}$
12. Monetary policy	$\hat{i}_t = \phi_\pi \hat{\pi}_t + e_{m,t}$
13. Phillips curve	$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{\kappa}{\sigma+\varphi}\widehat{mc}_t$

Aggregate Labor Supply

Back

- From individual labor supplies

$$N_{S,t}^{\varphi} C_{S,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{S,t}, \tag{31}$$

$$N_{H,t}^{\varphi} C_{H,t}^{\sigma} \exp(-\xi_t) = (1-\tau^w) W_{H,t}, \tag{32}$$

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- Using the Cobb-Douglas labor aggregator $N_t = N_{S,t}^{1-\lambda_t} N_{H,t}^{\lambda_t}$, we get:

$$N_t^{\varphi} = \exp(\xi_t)(1-\tau^w) W_t \left[C_{S,t}^{1-\lambda_t} C_{H,t}^{\lambda_t} \right]^{-\sigma}, \tag{33}$$

$$\tag{34}$$

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- Using the production function $Y_t = A_t N_t$, and $C_t = Y_t$ we get the aggregate labor supply:

$$(1-\tau^w) W_t = \exp(-\xi_t) A_t^{-\varphi} C_t^{\sigma+\varphi} Q_t. \tag{35}$$

Steady state

Back

- $\tau = \frac{1}{\epsilon}$, $\tau^w = 1 - Q$, $\mathcal{M} \equiv \frac{\epsilon}{\epsilon - 1}$.

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- $D = \frac{\delta}{\epsilon} Y > 0$ and $\tau^D < \lambda$.

$$\Gamma \equiv \frac{C_S}{C_H} \Rightarrow \Gamma = \frac{1 - \lambda + \lambda \frac{\delta}{\epsilon}\left(1 - \frac{\tau^D}{\lambda}\right)}{1 - \lambda - (1 - \lambda)\frac{\delta}{\epsilon}\left(1 - \frac{\tau^D}{\lambda}\right)} > 1. \tag{36}$$

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- Given $\tau^W = 1 - Q$, then $W = N^{\varphi} Y^{\sigma}$. Finally $\Gamma > 1 \Rightarrow Q_t < 1$ which implies that:

$$MC = \frac{1 - \tau}{1 - \tau^W} \frac{Y^{\sigma + \varphi}}{A^{1+\sigma}} Q \Rightarrow Y = A^{\frac{1+\sigma}{\sigma+\varphi}} = Y^*. \tag{37}$$

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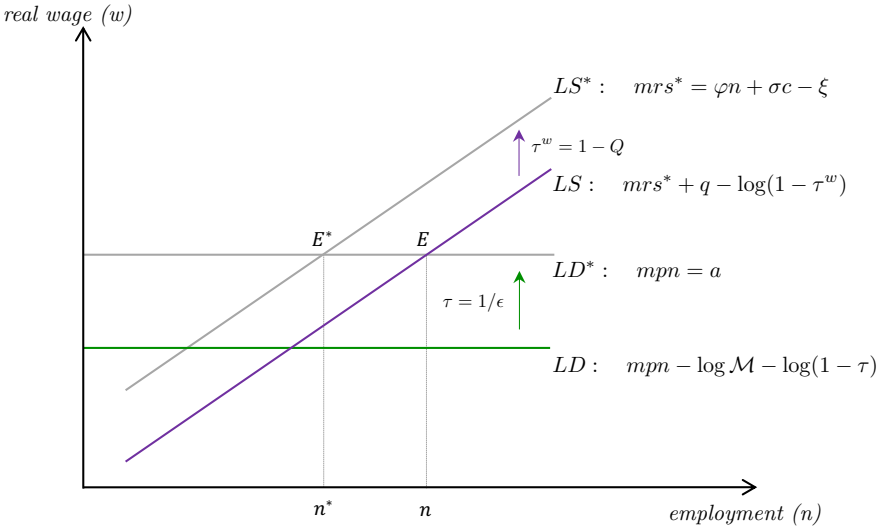
- Note: without the labor income tax, i.e. $\tau^W = 0$, $Y > Y^*$.

Steady-State



Back

Log-linearized Equilibrium Conditions

Back

Condition	Equation
Euler equation S	$\hat{c}_{S,t} = E_t \hat{c}_{S,t+1} - (1-p) E_t \hat{Y}_{t+1} - \frac{1}{\sigma}\left(\hat{\imath}_t - E_t \hat{\pi}_{t+1} + E_t \Delta \hat{\xi}_{t+1}\right) + p \frac{\Gamma^{\sigma} - 1}{\sigma} E_t \hat{p}_{S,t+1}$
Budget constraint H	$\hat{c}_{H,t} = \left(1 - \frac{\tau^D \delta}{\psi \epsilon}\right)(\hat{w}_t + \hat{n}_t) + \frac{\tau^D}{\psi} \hat{d}_t - \frac{\tau^D \delta}{\psi \epsilon} \hat{\lambda}_t$
Labor supply	$\hat{w}_t = -\hat{\xi}_t + \varphi \hat{n}_t + \sigma \hat{c}_t + \hat{q}_t$
Marginal costs	$\widehat{mc}_t = \hat{w}_t - \hat{a}_t$
Q_t wedge	$\hat{q}_t = \sigma(1-\lambda)\hat{c}_{S,t} + \sigma\lambda \hat{c}_{H,t} - \sigma \hat{y}_t - \sigma\gamma\lambda \hat{\lambda}_t$
Consump. inequality	$\hat{Y}_t = \hat{c}_{S,t} - \hat{c}_{H,t}$
Production function	$\hat{y}_t = \hat{a}_t + \hat{n}_t$
Aggregate consump.	$\hat{c}_t = (1-\psi)\hat{c}_{S,t} + \psi \hat{c}_{H,t} - \psi g \hat{\lambda}_t$

Log-linearized Equilibrium Conditions

Back

Condition	Equation
Resource constraint	$\hat{y}_t = \hat{c}_t$
Profits	$\hat{d}_t = \hat{y}_t - (1-\delta\tau)(\hat{w}_t + \hat{n}_t)$
Share dynamics	$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} - \eta\hat{y}_t + p_H e_{H,t} - \frac{1-\lambda}{\lambda} p_S e_{S,t}$
NK Phillips curve	$\hat{\pi}_t = \beta E_t \hat{\pi}_{t+1} + \frac{\kappa}{\sigma+\varphi}\widehat{mc}_t$
Persistence prob. S	$\hat{p}_{S,t} = \eta_S \hat{y}_t + e_{S,t}$
Persistence prob. H	$\hat{p}_{H,t} = -\eta_H \hat{y}_t + e_{H,t}$

Definitions of composite parameters

$$p \equiv \frac{p_S}{p_S + (1-p_S)\Gamma^{\sigma}}, \quad \psi \equiv \frac{\lambda}{\lambda + (1-\lambda)\Gamma}, \quad \rho \equiv p_S + p_H - 1 = 1 - \frac{1-p_S}{\lambda}, \quad \eta \equiv \frac{1-\lambda}{\lambda} p_S \eta_S + p_H \eta_H$$

Log-linear T-HANK

Partial profits redistribution case: $\tau^D = \psi < \lambda$ back

- $\hat{c}_{h,t} = \hat{y}_t + \frac{\delta}{\epsilon}\hat{\lambda}_t \longleftrightarrow \chi = 1$
- We are shutting down the amplification coming from countercyclical consumption inequality.

Description	Equation
IS curve	$x_t = E_t x_{t+1} - \frac{1}{\sigma}(\hat{i}_t - E_t\hat{\pi}_{t+1} - r_t^*) - \left[\theta(1-p\rho) + \rho(1-p)\frac{\delta}{\epsilon}\right]\hat{\lambda}_t + p\frac{\Gamma^{\sigma}-1}{\sigma}E_t\hat{p}_{S,t+1}$
NKPC	$\hat{\pi}_t = \beta E_t\hat{\pi}_{t+1} + \kappa x_t - \kappa\omega\hat{\lambda}_t$
Cons. inequality	$\hat{\gamma}_t = \frac{\varpi}{1-\psi}\hat{\lambda}_t$
Monetary policy	$\hat{i}_t = \phi_\pi\hat{\pi}_t + e_{m,t}$
H share	$\hat{\lambda}_t = \rho\hat{\lambda}_{t-1} + p_H e_{H,t} - \frac{1-\lambda}{\lambda}p_S e_{S,t}$
Natural rate	$\hat{r}_t^* \equiv \sigma E_t\Delta\hat{y}_{t+1}^* - E_t\Delta\hat{\xi}_{t+1}$
Potential output	$\hat{y}_t^* = \frac{1+\varphi}{\sigma+\varphi}\hat{a}_t + \frac{1}{\sigma+\varphi}\hat{\xi}_t$

Composite parameters

Composite parameters ($\tau^D = \psi < \lambda$)

Back

Composite parameters
$p_\gamma \equiv p_S + (1-p_S)\Gamma^\sigma$; $p \equiv \frac{p_S}{p_\gamma}$; $\psi \equiv \frac{\lambda}{\lambda+(1-\lambda)\Gamma} = \frac{\lambda}{1+g(1-\lambda)}$; $\chi = 1$; $\phi = 0$; $\Phi = 1$; $\sigma_x = \sigma$; $\omega_x = 0$; $\varpi = \frac{\delta}{\epsilon} + \psi g$; $\zeta = \psi\left(g + \frac{\delta}{\epsilon}\right)$; $\theta = \frac{\psi}{1-\psi}\left(g + \frac{\delta}{\epsilon}\right)$. $\nu = \frac{\psi\left(g+\frac{\delta}{\epsilon}\right)-(1-p)\left(\frac{\delta}{\epsilon}+\psi g\right)}{1-\psi}$; $\omega = \frac{\sigma}{\sigma+\varphi}\left[\frac{\lambda-\psi}{1-\psi}\left(\frac{\delta}{\epsilon} + \psi g\right) + \lambda\gamma - \psi g\right]$; $\rho = 1 - \frac{1-p_S}{\lambda} \in (0, 1)$.

Partial Equilibrium - Labor Market back

Employment

- Start from:

$$y_t^* = \frac{1+\varphi}{\sigma+\varphi} a_t + \frac{1}{\sigma+\varphi} \hat{\xi}_t$$

so

$$y_t^* - a = \frac{1+\varphi}{\sigma+\varphi} a_t - a + \frac{1}{\sigma+\varphi} \hat{\xi}_t.$$

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- Add and subtract $\frac{1+\varphi}{\sigma+\varphi}a$:

$$y_t^* - a = \underbrace{\left(\frac{1+\varphi}{\sigma+\varphi}a - a\right)}_{y^*-a} + \frac{1+\varphi}{\sigma+\varphi}(a_t - a) + \frac{1}{\sigma+\varphi}\hat{\xi}_t.$$

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- Assumption needed for the simplified expression: $a_t = a$ (no productivity deviation). Then

$$y_t^* - a = y^* - a + \frac{1}{\sigma+\varphi}\hat{\xi}_t, \qquad y^* - a = \frac{1-\sigma}{\sigma+\varphi}a.$$

Partial Equilibrium - Labor Market

Labor Supply back

- Start from the wage equation and subtract a:

$$w_t - a = -\hat{\xi}_t + \varphi n_t + \sigma y_t^* - a + \hat{q}_t$$

Partial Equilibrium - Labor Market

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we can rewrite

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Partial Equilibrium - Labor Market

Labor Supply back

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- Therefore,

$$w_t - a = \sigma y^* - a - \frac{\varphi}{\sigma + \varphi}\hat{\xi}_t + \varphi n_t + \hat{q}_t$$